

East Java Precipitation Anomalies Across Historical El Niño Episodes (2001–2025)

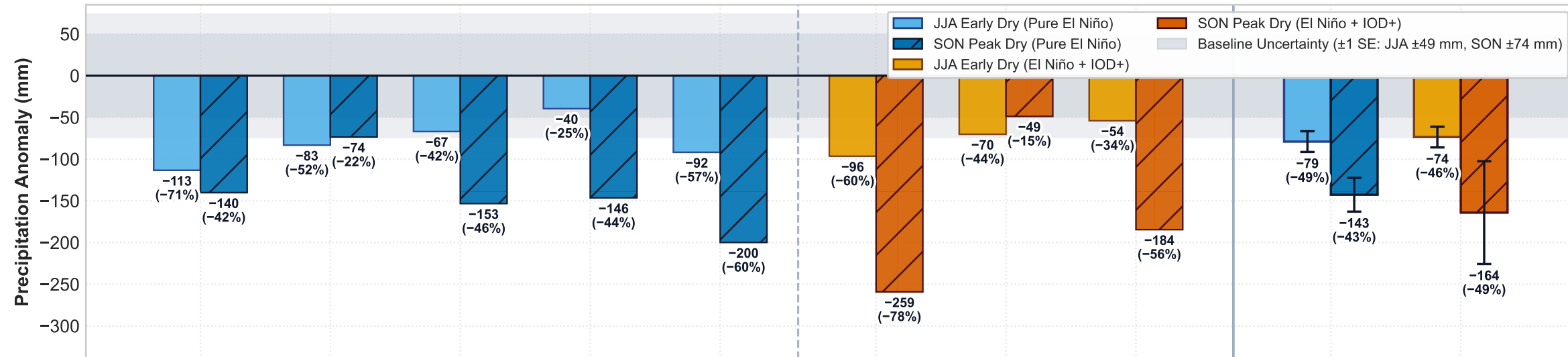
ENSO-Neutral Baseline Climatology (n = 6): JJA = 161 mm, SON = 332 mm | Data: UCSB CHIRPS v2.0 (0.05° resolution)

Pure El Niño (n = 5)

Compound: El Niño + IOD+ (n = 3)

Cohort Composites (±1 SEM)

(a) Seasonal Precipitation Anomaly (mm) and Relative Departure (% of Climatological Baseline)



(b) Standardized Meteorological Anomaly (Z-score relative to ENSO-Neutral Baseline)

